Assignment 6

1. Display manager ID and number of employees managed by the manager.

```
D6_99024_tejas>select MANAGER_ID , count(EMPLOYEE_ID)
    -> from employees
    -> group by manager_ID;
+------------+--------------------+
| MANAGER_ID | count(EMPLOYEE_ID) |
+------------+--------------------+
|       NULL |                  1 |
|        100 |                 14 |
|        101 |                  5 |
|        102 |                  1 |
|        103 |                  4 |
|        108 |                  5 |
|        114 |                  5 |
|        120 |                  8 |
|        121 |                  8 |
|        122 |                  8 |
|        123 |                  8 |
|        124 |                  8 |
|        145 |                  6 |
|        146 |                  6 |
|        147 |                  6 |
|        148 |                  6 |
|        149 |                  6 |
|        201 |                  1 |
|        205 |                  1 |
+------------+--------------------+
```

2. Display the country ID and number of cities we have in the country.

```
D6_99024_tejas>select COUNTRY_ID,count(city
    -> from locations
    -> group by COUNTRY_ID;
+------------+-------------+
| COUNTRY_ID | count(city) |
+------------+-------------+
| AU         |           1 |
| BR         |           1 |
| CA         |           2 |
| CH         |           2 |
| CN         |           1 |
| DE         |           1 |
| GB         |           3 |
| IN         |           1 |
| IT         |           2 |
| JP         |           2 |
| MX         |           1 |
| NL         |           1 |
| SG         |           1 |
| US         |           4 |
+------------+-------------+
```

3.Display average salary of employees in each department who have commission percentage.

```
D6_99024_tejas>select department_id , avg(salary)
    -> from employees
    -> group by department_id;
+---------------+--------------+
| department_id | avg(salary)  |
+---------------+--------------+
|          NULL |  7000.000000 |
|            10 |  4400.000000 |
|            20 |  9500.000000 |
|            30 |  4150.000000 |
|            40 |  6500.000000 |
|            50 |  3475.555556 |
|            60 |  5760.000000 |
|            70 | 10000.000000 |
|            80 |  8955.882353 |
|            90 | 19333.333333 |
|           100 |  8601.333333 |
|           110 | 10154.000000 |
+---------------+--------------+
```

4. Display job ID, number of employees, sum of salary, and difference between highest salary and lowest salary of the employees for each job.

```
   -> select JOB_ID,count(EMPLOYEE_ID),sum(salary),max(salary)-min(salary) as diff
   ->     from employees
   ->     group by JOB_ID;
+------------+--------------------+-------------+---------+
| JOB_ID     | count(EMPLOYEE_ID) | sum(salary) | diff    |
+------------+--------------------+-------------+---------+
| AC_ACCOUNT |                  1 |     8300.00 |    0.00 |
| AC_MGR     |                  1 |    12008.00 |    0.00 |
| AD_ASST    |                  1 |     4400.00 |    0.00 |
| AD_PRES    |                  1 |    24000.00 |    0.00 |
| AD_VP      |                  2 |    34000.00 |    0.00 |
| FI_ACCOUNT |                  5 |    39600.00 | 2100.00 |
| FI_MGR     |                  1 |    12008.00 |    0.00 |
| HR_REP     |                  1 |     6500.00 |    0.00 |
| IT_PROG    |                  5 |    28800.00 | 4800.00 |
| MK_MAN     |                  1 |    13000.00 |    0.00 |
| MK_REP     |                  1 |     6000.00 |    0.00 |
| PR_REP     |                  1 |    10000.00 |    0.00 |
| PU_CLERK   |                  5 |    13900.00 |  600.00 |
| PU_MAN     |                  1 |    11000.00 |    0.00 |
| SA_MAN     |                  5 |    61000.00 | 3500.00 |
| SA_REP     |                 30 |   250500.00 | 5400.00 |
| SH_CLERK   |                 20 |    64300.00 | 1700.00 |
| ST_CLERK   |                 20 |    55700.00 | 1500.00 |
| ST_MAN     |                  5 |    36400.00 | 2400.00 |
+------------+--------------------+-------------+---------+
```

5. Display job ID for jobs with average salary more than 10000.

```
D6_99024_tejas>select JOB_ID ,avg(salary)
    -> from employees
    -> group by JOB_ID
    -> having avg(salary)>10000;
+---------+--------------+
| JOB_ID  | avg(salary)  |
+---------+--------------+
| AC_MGR  | 12008.000000 |
| AD_PRES | 24000.000000 |
| AD_VP   | 17000.000000 |
| FI_MGR  | 12008.000000 |
| MK_MAN  | 13000.000000 |
| PU_MAN  | 11000.000000 |
| SA_MAN  | 12200.000000 |
+---------+--------------+
```

6. Display the years in which more than 10 employees joined.

```
D6_99024_tejas>select year(hire_date) AS HIRE_DATE,count(employee_id) AS EMP_ID
    -> from employees
    -> GROUP BY year(hire_date)
    -> HAVING count(employee_id) > 10;
+-----------+--------+
| HIRE_DATE | EMP_ID |
+-----------+--------+
|      2015 |     29 |
|      2016 |     24 |
|      2017 |     19 |
|      2018 |     11 |
+-----------+--------+
```

7. Display departments in which more than five employees have commission percentage.

```
D6_99024_tejas>SELECT DEPARTMENT_ID,COUNT(EMPLOYEE_ID)
    -> FROM EMPLOYEES
    -> WHERE COMMISSION_PCT > 0.05
    -> GROUP BY DEPARTMENT_ID;
+---------------+--------------------+
| DEPARTMENT_ID | COUNT(EMPLOYEE_ID) |
+---------------+--------------------+
|          NULL |                  1 |
|            80 |                 34 |
+---------------+--------------------+
```

8. Display employee ID for the employees who did more than one job in the past.

```
D6_99024_tejas>select EMPLOYEE_ID ,count(job_id)
    -> from job_history
    -> group by employee_id
    -> having count(job_id)>1;
+-------------+---------------+
| EMPLOYEE_ID | count(job_id) |
+-------------+---------------+
|         101 |             2 |
|         176 |             2 |
|         200 |             2 |
+-------------+---------------+
```

9. Display job ID of jobs that were done by 2 or more employees for more than 100 days individually.

```
D6_99024_tejas>SELECT job_id , count(DISTINCT employee_id)
    -> FROM job_history
    -> WHERE TIMESTAMPDIFF(DAY, start_date, end_date) > 100
    -> GROUP BY job_id
    -> HAVING COUNT(DISTINCT employee_id) >= 2;
+------------+-----------------------------+
| job_id     | count(DISTINCT employee_id) |
+------------+-----------------------------+
| AC_ACCOUNT |                           2 |
| ST_CLERK   |                           2 |
+------------+-----------------------------+
```

10. Display the department ID, year of hiring, and the number of employees who joined each department in each year.

```
D6_99024_tejas>select department_id,YEAR(hire_date) ,count(employee_id)
    -> from employees
    -> group by department_id,YEAR(hire_date)
    -> order by department_id,year(hire_date);
+---------------+-----------------+--------------------+
| department_id | YEAR(hire_date) | count(employee_id) |
+---------------+-----------------+--------------------+
|          NULL |            2017 |                  1 |
|            10 |            2013 |                  1 |
|            20 |            2014 |                  1 |
|            20 |            2015 |                  1 |
|            30 |            2012 |                  1 |
|            30 |            2013 |                  1 |
|            30 |            2015 |                  2 |
|            30 |            2016 |                  1 |
|            30 |            2017 |                  1 |
|            40 |            2012 |                  1 |
|            50 |            2013 |                  3 |
|            50 |            2014 |                  4 |
|            50 |            2015 |                 12 |
|            50 |            2016 |                 13 |
|            50 |            2017 |                  9 |
|            50 |            2018 |                  4 |
|            60 |            2015 |                  1 |
|            60 |            2016 |                  2 |
|            60 |            2017 |                  2 |
|            70 |            2012 |                  1 |
|            80 |            2014 |                  5 |
|            80 |            2015 |                 10 |
|            80 |            2016 |                  7 |
|            80 |            2017 |                  5 |
|            80 |            2018 |                  7 |
|            90 |            2011 |                  1 |
|            90 |            2013 |                  1 |
|            90 |            2015 |                  1 |
|           100 |            2012 |                  2 |
|           100 |            2015 |                  2 |
|           100 |            2016 |                  1 |
|           100 |            2017 |                  1 |
|           110 |            2012 |                  2 |
+---------------+-----------------+--------------------+
```

11. Display the count of employees joined in each month of the year 2017.

```
D6_99024_tejas>SELECT MONTH(HIRE_DATE) AS MONTH, COUNT(EMPLOYEE_ID) AS EMP_COUNT
    -> FROM EMPLOYEES
    -> WHERE YEAR(HIRE_DATE) = 2017
    -> GROUP BY MONTH(HIRE_DATE)
    -> ORDER BY MONTH(HIRE_DATE);
+-------+-----------+
| MONTH | EMP_COUNT |
+-------+-----------+
|     1 |         1 |
|     2 |         3 |
|     3 |         3 |
|     4 |         1 |
|     5 |         2 |
|     6 |         2 |
|     8 |         1 |
|    10 |         1 |
|    11 |         2 |
|    12 |         3 |
+-------+-----------+
```

12. Display the details of departments in which the maximum salary is more than 10000.

```
D6_99024_tejas>SELECT DEPARTMENT_ID, MAX(SALARY) AS MAX_SALARY
    -> FROM EMPLOYEES
    -> GROUP BY DEPARTMENT_ID
    -> HAVING MAX(SALARY) > 10000;
+---------------+------------+
| DEPARTMENT_ID | MAX_SALARY |
+---------------+------------+
|            20 |   13000.00 |
|            30 |   11000.00 |
|            80 |   14000.00 |
|            90 |   24000.00 |
|           100 |   12008.00 |
|           110 |   12008.00 |
+---------------+------------+
```

1. Write a query that counts the number of salespeople registering orders for each day. (If a salesperson has more than one order on a given day, he or she should be counted only once.).\

```
D6_99024_tejas>select odate, count(distinct snum) as salespeople_count
    -> from orders
    -> group by odate;
+------------+-------------------+
| odate      | salespeople_count |
+------------+-------------------+
| 1990-10-03 |                 4 |
| 1990-10-04 |                 3 |
+------------+-------------------+
```

3 Write an SQL query to calculate the total order amount for each day and display the results in descending order of total orders.

```
D6_99024_tejas>select odate, sum(amt) as total_order_amount
    -> from orders
    -> group by odate
    -> order by total_order_amount desc;
+------------+--------------------+
| odate      | total_order_amount |
+------------+--------------------+
| 1990-10-04 |           16713.81 |
| 1990-10-03 |            8944.59 |
+------------+--------------------+
```

4. Write a query that selects the total amount in orders for each salesperson for whom this total is greater than the average amount of the order in the table. (Note Use the average amount value directly →2565.84)

```
D6_99024_tejas>select snum, sum(amt) as total_amount
    -> from orders
    -> group by snum
    -> having sum(amt) > 2565.84;
+------+--------------+
| snum | total_amount |
+------+--------------+
| 1001 |     15382.07 |
| 1002 |      5546.15 |
+------+--------------+
```

5. Write a query that selects the highest rating in each city.

```
D6_99024_tejas>select city, max(rating) as highest_rating
    -> from customers
    -> group by city;
+----------+----------------+
| city     | highest_rating |
+----------+----------------+
| London   |            100 |
| Rome     |            200 |
| San Jose |            300 |
| Berlin   |            300 |
+----------+----------------+
```